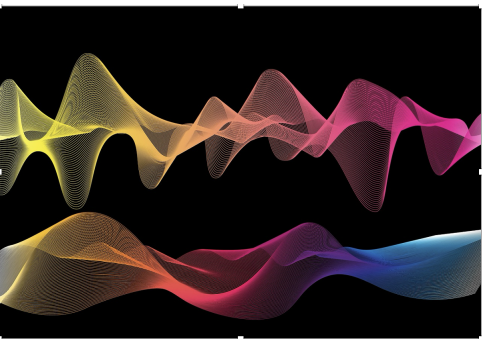


Types of Data

Procheta Sen



Two General Classes of Data

- ▶ Nondependency-oriented data: objects do not have dependencies
- ▶ Dependency-oriented data: implicit or explicit dependencies between objects may exist
 - ▶ Networks: nodes (objects) are connected by edges (relationships)
 - ▶ Successive measurements collected from a sensor

Nondependency-oriented data (multidimensional data)

- ▶ The simplest form of data.
- ▶ A multidimensional data set D typically contains a set of records $X_1, X_2 \dots X_n$.
- ▶ Each record X_i containing a set of d features $(x_i^1, x_i^2 \dots x_i^d)$.
- ▶ This data set can be represented by a $n * d$ data matrix.

Types of Data

- ▶ Numerical or quantitative (values have natural ordering)
 - ▶ Integer values (number of petals in a flower)
 - ▶ Real values (length of a petal)
- ▶ Categorical or unordered discrete-value
 - ▶ Discrete unordered values/categories (colour of a flower petal)
- ▶ Binary data (two values: 0 and 1)
 - ▶ Can be seen as a categorical data (two categories) or a numerical data (0/1)
 - ▶ Can be used to represent Set Data via characteristic vectors
- ▶ Text data
 - ▶ Document as a string (dependency-oriented data type)
 - ▶ Document as a set of words or terms (vector-space representation: frequencies of the words in the document)

Dependency-Oriented Data

- ▶ Implicit dependencies
 - ▶ Are not explicitly specified but are known to exist
 - ▶ Example: temperature values collected by a sensor
- ▶ Explicit dependencies
 - ▶ Graphs or network data (edges specify explicit relationships)

Types of Data with Implicit Dependencies

- ▶ Time-series
 - ▶ values that are generated by sequential measurements over time (time-stamp or index value is a contextual attribute; the measurement is behavioral attribute)
- ▶ Discrete Sequences and Strings
 - ▶ The categorical analog of time-series data
- ▶ Spatial data (every record has a location attribute)
 - ▶ Example: temperature, pressure are measured at spacial locations
- ▶ Spatiotemporal data (contain both spatial and temporal attributes)

Types of Data with Explicit Dependencies

- ▶ Network/Graph data
 - ▶ Objects correspond to nodes of the network.
 - ▶ Relationships between the objects correspond to the edges of the network.
 - ▶ Edges may be directed or undirected.
 - ▶ A set of attributes may be associated with a node.
 - ▶ A set of attributes may be associated with an edge.
 - ▶ Examples:
 - ▶ Web graph
 - ▶ Facebook/Instagram/LinkedIn social networks